$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

In other words,

$$(\sigma_z \otimes I)|uu\rangle = |uu\rangle,$$

just as we expect. What if we apply the same matrix to the column vector |*du*> in Eqs. 7.5? Carrying out the matrix multiplication results in −|*du*>, just as it should.

7.1.2 Building Tensor Product Matrices from Component Matrices

The above method for calculating matrix elements is very general—it works for *all* observables. If we need to construct the tensor product of two operators, and we *already know* the matrix elements of the building blocks, we can combine them directly. Here is the rule for combining 2 × 2 matrices to form 4 × 4 matrices:

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B \\ A_{21}B & A_{22}B \end{pmatrix} \qquad (7.6)$$

or

$$A \otimes B = \begin{pmatrix} A_{11}B_{11} & A_{11}B_{12} & A_{12}B_{11} & A_{12}B_{12} \\ A_{11}B_{21} & A_{11}B_{22} & A_{12}B_{21} & A_{12}B_{22} \\ A_{21}B_{11} & A_{21}B_{12} & A_{22}B_{11} & A_{22}B_{12} \\ A_{21}B_{21} & A_{21}B_{22} & A_{22}B_{21} & A_{22}B_{22} \end{pmatrix}. \qquad (7.7)$$